Quantum Mechanics Wave Function Problems

Formatted Problem Set

Questions generated below. Multiple-choice answers in RED; incorrect answers in BLUE.

Q1. A particle of mass m is confined to the region $0 < x < a$ and at $t = 0$ has wave function $\psi(x) = A\,x(a - x)$ for $0 < x < a$, with $\psi(x) = 0$ elsewhere.

(a) Find the normalization constant A.

Answer: *Normalization requires* $\int_0^a |\psi(x)|^2\,dx = 1$, *so* $A^2 \int_0^a x^2(a-x)^2\,dx = 1$. *Since* $\int_0^a x^2(a-x)^2\,dx = \frac{a^5}{30}$, *the constant is* $A = \sqrt{\frac{30}{a^5}}$.

(b) What equation should be used to compute the expectation value $\langle x \rangle$? Evaluate it for this state.

Answer: *Use* $\langle x \rangle = \int_{-\infty}^{\infty} \psi^*(x)\,x\,\psi(x)\,dx$, *which here reduces to* $\langle x \rangle = \int_0^a x|\psi(x)|^2\,dx$. *Because* $|\psi(x)|^2 = A^2x^2(a-x)^2$ *is symmetric about* $x = \frac{a}{2}$, *the result is* $\langle x \rangle = \frac{a}{2}$.

Q2. At time $t = 0$, a particle on the half-line $x \geq 0$ is described by $\psi(x, 0) = \sqrt{\frac{2}{b}}\,e^{-x/b}$ for $x \geq 0$, where $b > 0$, and $\psi(x, 0) = 0$ for $x < 0$. What is the probability of finding the particle to the left of $x = b$?

Answer: *The probability is* $P(x < b) = \int_0^b |\psi(x,0)|^2\,dx = \int_0^b \frac{2}{b}e^{-2x/b}\,dx$. *Evaluating the integral gives* $P(x < b) = 1 - e^{-2}$.

Q3. Inside a one-dimensional box of length L, consider the wave function $\psi(x, t) = \sqrt{\frac{2}{L}}\sin\left(\frac{3\pi x}{L}\right)e^{-iE_3t/\hbar}$ for $0 < x < L$, with $\psi = 0$ outside the box. Why is this a stationary state? Explain your answer using the probability density.

Answer: *This is a stationary state because the time dependence appears only as the phase factor* $e^{-iE_3t/\hbar}$, *multiplying a single spatial eigenfunction. Therefore* $|\psi(x,t)|^2 = \frac{2}{L}\sin^2\left(\frac{3\pi x}{L}\right)$, *which has no time dependence, so all position probabilities remain constant in time.*

Q4. A particle of mass m is trapped in a one-dimensional rigid box with walls at $y = 0$ and $y = a$. The potential is $V(y) = 0$ for $0 < y < a$ and $V(y) = \infty$ elsewhere.

(a) Write the time-independent Schrödinger equation inside the box and give its general solution in the allowed region.

Answer: *For* $0 < y < a$, *the potential is zero, so the equation is* $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dy^2} = E\psi$, *or equivalently* $\psi'' + k^2\psi = 0$ *with* $k^2 = \frac{2mE}{\hbar^2}$. *The general solution inside the box is* $\psi(y) = C\sin(ky) + D\cos(ky)$, *while* $\psi = 0$ *outside because the walls are infinite.*

(b) Apply the boundary conditions at the walls to determine the allowed energies.

Answer: *The boundary conditions are* $\psi(0) = 0$ *and* $\psi(a) = 0$. *From* $\psi(0) = 0$ *we get* $D = 0$, *and from* $\psi(a) = 0$ *we require* $\sin(ka) = 0$, *so* $ka = n\pi$ *with* $n = 1, 2, 3, \ldots$. *Hence* $E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}$.

(c) Find the normalized energy eigenfunctions.

Answer: *Using* $k_n = \frac{n\pi}{a}$, *the eigenfunctions inside the box are* $\psi_n(y) = N\sin\left(\frac{n\pi y}{a}\right)$. *Normalization* $\int_0^a |\psi_n(y)|^2\,dy = 1$ *gives* $N = \sqrt{\frac{2}{a}}$, *so* $\psi_n(y) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi y}{a}\right)$ *for* $0 < y < a$, *and zero elsewhere.*

(d) Show that ψ_n and ψ_m are orthogonal when $n \neq m$.

Answer: *For* $n \neq m$, $\int_0^a \psi_n(y)\psi_m(y)\,dy = \frac{2}{a}\int_0^a \sin\left(\frac{n\pi y}{a}\right)\sin\left(\frac{m\pi y}{a}\right)dy$. *This standard sine integral is zero when* $n \neq m$, *so* $\langle n|m\rangle = 0$ *and the eigenfunctions are orthogonal.*

Q5. A particle in a one-dimensional infinite square well of width L has initial state $\Psi(x,0) = \frac{\sqrt{3}}{2}\,\psi_1(x) + \frac{1}{2}\,\psi_3(x)$, where ψ_n is the n-th stationary state of the well. Find the expectation value of the particle's energy.

Answer: *Since the coefficients are $c_1 = \frac{\sqrt{3}}{2}$ and $c_3 = \frac{1}{2}$, the probabilities are $|c_1|^2 = \frac{3}{4}$ and $|c_3|^2 = \frac{1}{4}$. Therefore $\langle E\rangle = \frac{3}{4}E_1 + \frac{1}{4}E_3 = \frac{3}{4}E_1 + \frac{9}{4}E_1 = 3E_1 = \frac{3\pi^2\hbar^2}{2mL^2}$.*
